

Pavan Tiwari

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Professional Summary

Senior Applied AI Engineer with **5+ years** of experience architecting and deploying production-grade **Machine Learning (ML)**, **Deep Learning (DL)**, and **Generative AI (GenAI)** systems across manufacturing and finance domains. Generated **170 Cr business impact** through high-frequency time-series forecasting; eliminated **10+ FTE** of manual effort via end-to-end intelligent automation. Proven expertise in **LLM/RAG** architectures, **Agentic AI** (LangChain, LangGraph, OpenAI SDK, MCP), and scalable MLOps on AWS.

Experience

Senior Data Scientist / Applied AI Engineer

AM/NS India — ArcelorMittal Nippon Steel India

Apr 2021 – Present

Navsari, Gujarat, India

- Spearheaded **GenAI Document Intelligence** platform in **Python** using LangChain, LangGraph, and RAG (FAISS/ChromaDB); semantic chunking, agentic retrieval loops, and LLM-based QA over enterprise knowledge bases — slashed manual analysis by **60%**.
- Developed **Iron Procurement Multi-step Agentic System** in **Python** (OpenAI SDK, MCP, FastAPI) — 6-step ReAct agent automating PDF parsing, 3-channel supplier evaluation, and split-order recommendations via Excel & HTML email.
- Engineered **EAF Power Consumption Forecasting System** in **Python** — XGBoost, Random Forest, and LSTM (TensorFlow) ensemble predicting 12×15-min MW blocks from 5-sec sensor data on AWS RDS; generated **170 Cr profit optimization in FY2025** and eliminated **7 FTE** of manual workload.
- Architected **Treasury Inflow Forecasting System** in **Python** across 5 financial streams using ARIMA, 2-state HMM, and two-stage XGBoost (Scikit-learn); delivered P10/P50/P90 forecasts, enhanced cash-flow visibility, and eliminated **3 FTE** of manual effort.

Data Science Intern

Techno Labs

Oct 2020 – Mar 2021

India

- Applied supervised and unsupervised ML models to surface data patterns; built interactive Python dashboards and delivered data-driven insights to client stakeholders.

Technical Skills

Languages & DB	Python, SQL, PostgreSQL, AWS RDS
ML / Ensemble	XGBoost, Random Forest, LightGBM, Scikit-learn, HMM
Deep Learning	LSTM, CNN, Neural Networks, TensorFlow
Time Series	ARIMA, SARIMAX, Prophet, ML-based forecasting
GenAI / LLM	LangChain, LangGraph, OpenAI SDK, Claude API, Hugging Face, LlamaIndex
Agentic AI	ReAct, MCP, Tool Calling, Multi-agent Systems, CrewAI
Vector Databases	FAISS, Pinecone, ChromaDB, AstraDB
Cloud & MLOps	AWS (RDS, EC2, S3), MLflow, Drift Detection, Automated Retraining
DevOps	Docker, Kubernetes
Tools	FastAPI, React, Git, GitHub, Jupyter Notebook, VS Code, PyCharm, N8N Automation

Key Projects

Iron Procurement Agentic AI | *Python, Agentic AI, OpenAI SDK, MCP, LangChain, LangGraph, FastAPI*

- 6-step ReAct multi-agent system for PDF parsing, 3-channel supplier comparison, and split-order recommendations via Excel + HTML email.

GenAI Document Intelligence with Agentic RAG | *Python, Generative AI, RAG, LangChain, LangGraph, FAISS, ChromaDB*

- LangGraph-orchestrated agentic RAG with semantic chunking and LLM-based QA over enterprise knowledge bases; reduced manual document analysis by **60%**.

EAF Power Bidding & Forecasting System | *Python, Machine Learning, Deep Learning, XGBoost, Random Forest, LSTM,*

TensorFlow, AWS

- Production ensemble forecaster with 30-min scheduling, weekly retraining, strict leakage prevention (temporal splits, train-only scaler), and self-healing PKL versioning; core driver of **170 Cr** FY2025 impact.

Treasury Inflow Forecasting System | *Python, Machine Learning, ARIMA, HMM, XGBoost, Scikit-learn, SMTP*

- OOP 5-component forecaster with 2-state HMM regime detection, P10/P50/P90 quantile bands, dual-channel SMTP reporting, and self-healing model versioning; eliminated **3 FTE** of manual effort.

Surface Defect Detection System (SDDS) | *Python, Deep Learning, CNN, TensorFlow, Transfer Learning, Scikit-learn*

- Transfer learning-based vision classifier identifying **8 steel surface defect types** at **75% accuracy**; reduced inspection turnaround time and earned **Business Excellence Award**.

Education

PG Diploma in Artificial Intelligence & Machine Learning

Indian Institute of Technology (IIT), Guwahati

2020 – 2021

CGPA: 7.92 / 10

Bachelor of Technology, Computer Science & Engineering

Oriental University, Indore

2016 – 2020

CGPA: 8.1 / 10

Achievements & Awards

- Best Cost Saving Award** — Power Forecasting contributed to **170 Cr profit optimization** (FY2025); promoted for measurable business impact.
- Business Excellence Award** — SDDS improved steel quality inspection efficiency through deep learning-based automation.